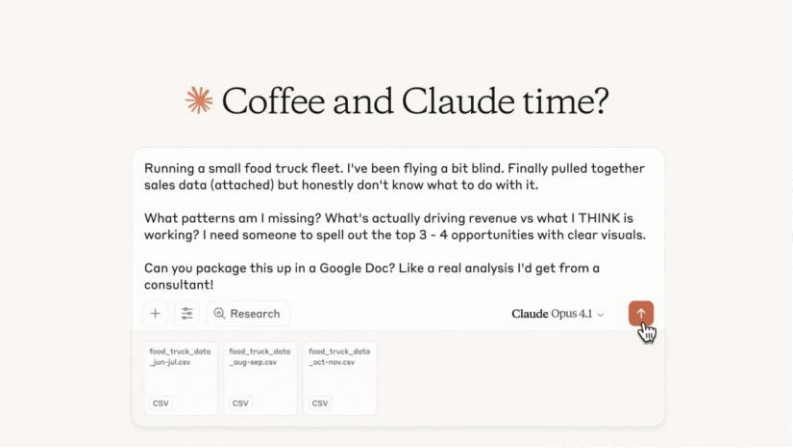
NIELIT Ropar & SCERT Punjab

Enhancing Digital Productivity

using Advanced Word & Excel



Day 1 · Session 1 | 9:30 AM – 11:15 AM

Teachers Training in ICT & Digital Initiatives

Claude AI is an advanced AI assistant developed by Anthropic. It is a large language model system designed for tasks like writing, coding, data analysis, and document generation

Session Agenda

01 **Word Productivity Essentials**
Styles, Navigation, Templates & Macros

02 **AI in Microsoft Word**
Copilot, Editor, Smart Suggestions & Automation

03 **Excel Advanced Techniques**
PivotTables, VLOOKUP/XLOOKUP, Data Validation & Charts

04 **AI in Microsoft Excel**
Analyze Data, Flash Fill, Copilot & Predictive Analytics

05 **Hands-On Practicals**
Word Practical + Excel Practical with AI Integration

Why Digital Productivity Matters

73%

Teachers spend extra time on repetitive document tasks

3x

Faster document creation with advanced formatting tools

60%

Data analysis errors reduced using Excel functions

40%

Time saved with AI-assisted writing and analysis

Microsoft Word - Advanced Features

Styles & Formatting

- Heading styles (H1–H6) for auto-TOC
- Character vs paragraph styles
- Style sets & themes
- Format Painter shortcuts

Document Navigation

- Navigation Pane (Ctrl+F)
- Bookmarks & cross-references
- Document Map overview
- Go To (F5) shortcuts

Templates & Themes

- Create custom .dotx templates
- Default template modification
- Professional theme application
- Content placeholders & fields

Macros & Automation

- Record macros for repetitive tasks
- Assign macros to quick access toolbar
- Basic VBA introduction
- AutoText & Building Blocks

Microsoft Word - Advanced Features

Tables & Mail Merge

- Advanced table formatting
- Nested tables technique
- Mail Merge with Excel data source
- Merge fields & conditional merge

Review & Collaboration

- Track Changes with multiple authors
- Comments & threaded replies
- Compare & Combine documents
- Document protection & restrictions

References & Citations

- Table of Contents (auto-generate)
- Footnotes & Endnotes
- Bibliography & Citations
- Index & Table of Figures

Advanced Printing & Export

- Custom page sizes & margins
- Section breaks for mixed layouts
- PDF/XPS export with accessibility tags
- Print areas & custom headers/footers

AI in Microsoft Word

Microsoft Copilot & Smart AI Tools

✍ Copilot (claude.ai) — Draft & Rewrite

Generate entire documents from prompts. Rewrite sections in different tones — formal, casual, concise, or elaborate. Ask Copilot or claude.ai to summarize long documents.

Microsoft Editor

AI-powered grammar, clarity, and style checker. Suggests vocabulary improvements, conciseness edits, and formal tone adjustments in real-time.

Smart Lookup & Researcher

Right-click any word for AI-powered definitions and Wikipedia summaries. Researcher tool finds credible sources and inserts citations automatically.

Designer & Format Suggestions

AI suggests professional layout changes. Auto-formatting recommendations for headings, spacing, and consistency across the document.

Dictation & Transcribe

Voice-to-text with AI punctuation insertion. Transcribe audio recordings into formatted Word text with speaker identification.

Summarize & Transform

Copilot can summarize selected text or entire document. Transform bullet points into prose or tables. Create PowerPoint outlines directly from Word docs.

Microsoft Excel - Advanced Features

Lookup Functions

- VLOOKUP / HLOOKUP for table data retrieval
- XLOOKUP — Next-gen lookup (any direction)
- INDEX + MATCH combination
- Dynamic array formulas with FILTER & SORT

PivotTables & Charts

- Create PivotTable from large datasets
- Grouping, filtering & slicers
- PivotChart creation & formatting
- Calculated fields & items

Advanced Formulas

- IF / IFS / IFERROR nesting
- SUMIFS, COUNTIFS, AVERAGEIFS
- TEXT functions (LEFT, MID, CONCATENATE)
- Date/Time functions for reporting

Data Validation & Protection

- Input restrictions (lists, numbers, dates)
- Custom validation rules
- Sheet & workbook protection
- Conditional formatting rules

AI in Microsoft Excel

Copilot, Flash Fill, Analyze Data & Predictive Analytics

⚡ Flash Fill

AI detects patterns in your data automatically. Enter a few examples — Excel fills the rest. Works for name splitting, phone number formatting, and text transformation without formulas.

📈 Analyze Data (Ideas)

Click 'Analyze Data' — AI scans your dataset and shows pre-built PivotTables, charts, and trends. Asks natural language questions about your data instantly.

Copilot in Excel

Type plain English queries: 'Show me total sales by region' — Copilot creates formulas, charts, and analyses. Highlights anomalies and patterns automatically.

🔮 Forecast Sheet

AI-powered time-series forecasting. Select sales/date data → Insert → Forecast Sheet. Excel predicts future values with confidence intervals automatically.

Data Types (AI-enriched)

Type company or country names → Mark as Data Type → Excel fetches live data (population, stock price, geography). Expands records dynamically.

✅ Smart Suggestions

Excel auto-suggests formulas as you type. AutoComplete for function arguments. Error-checking with one-click AI fix suggestions and formula explanations.

Practical 1 - Microsoft Word

Hands-On: Create a Professional School Circular / Notice

TASK: Create a formatted school circular for a Parent-Teacher Meeting using Word's advanced features

Step 1

Apply Heading Styles

Open a blank Word document. Apply Title style to school name, Heading 1 for 'Notice', Heading 2 for sections. Generate automatic Table of Contents.

Step 2

Insert Header & Footer

Add school logo (placeholder) in header. Insert page numbers in footer. Add auto-date field using Insert > Quick Parts > Field > Date.

Step 3

Create a Formatted Table

Insert a 5×4 table for meeting schedule (Class | Teacher | Time | Room). Apply Table Style, bold headers, alternate row shading.

Step 4

Mail Merge for Parents

Create an Excel sheet with Parent Name, Class, Child Name columns. Use Mailings > Start Mail Merge > Letters. Insert merge fields for personalization.

Step 5

Use AI — Word Copilot/Editor

Select the notice text → click Editor → accept grammar/style suggestions. Ask Copilot to 'Make this notice more formal' or 'Summarize in 3 bullet points'.

Step 6

Export & Share

Save as .docx template (.dotx). Export to PDF via File > Export. Use 'Share' to send a view-only link to colleagues for review.

Practical 2 - Microsoft Excel with AI

Hands-On: Student Marks Analysis Dashboard with AI Tools

TASK: Build a Student Marks Analysis sheet with XLOOKUP, PivotTable, Charts, and AI-powered insights

Step 1

Create Data Table

Set up columns: Student Name, Class, Sub1, Sub2, Sub3, Sub4, Sub5, Total, Average, Grade. Enter data for 20 students. Format as Excel Table (Ctrl+T).

Step 2

Formulas & Functions

Use SUM for totals. Use AVERAGE for average. Use IFS for grade: A (≥90), B (≥75), C (≥60), D (≥45), F (<45). Use RANK to add student ranking.

Step 3

XLOOKUP — Search System

Create a 'Search by Name' box. Use XLOOKUP to pull a student's full record. Show 'Student not found' if name doesn't exist using [if_not_found] parameter.

Step 4

Flash Fill AI Demo

In a new column, type formatted output (e.g., 'Rahul - Class 9A'). Press Ctrl+E to trigger Flash Fill. Excel AI auto-fills all rows by detecting the pattern.

Step 5

PivotTable & Charts

Insert PivotTable → Class vs Average Marks. Add a Slicer for Class filter. Create Clustered Bar Chart. Add a Sparkline column for mini trends per student.

Step 6

Analyze Data (AI) + Forecast

Click 'Analyze Data' — AI shows instant insights. Then select 3-month trend data → Insert → Forecast Sheet. See AI-predicted future performance with confidence bands.

Essential Shortcuts Cheat Sheet

Word & Excel — Time-Saving Keyboard Shortcuts for Teachers

WORD Shortcuts

Ctrl + Alt + 1/2/3	Apply Heading 1/2/3
Ctrl + Shift + N	Apply Normal style
F12	Save As dialog
Ctrl + F	Open Navigation Pane
Alt + Shift + D	Insert current date
Ctrl + Shift + C/V	Copy/Paste formatting
Alt + F9	Toggle field codes
Ctrl + Enter	Insert page break

EXCEL Shortcuts

Ctrl + T	Create table from range
Ctrl + E	Flash Fill (AI pattern)
Alt + =	AutoSum selected cells
Ctrl + Shift + L	Toggle filters
F4	Toggle absolute reference ($)
Ctrl + ;	Insert today's date
Ctrl + `	Show all formulas
Alt + F1	Insert chart instantly

Session Summary & Key Takeaways

- Advanced Word skills — Styles, Templates, Mail Merge & Collaboration tools save hours of manual formatting work
- AI in Word (Copilot + Editor) can draft, rewrite, summarize, and improve your documents in seconds
- Excel functions (XLOOKUP, SUMIFS, PivotTables) transform raw data into meaningful reports automatically
- Flash Fill, Analyze Data, and Copilot in Excel replace manual analysis with AI-powered instant insights
- Both tools together enable teachers to create professional documents and analyze student data with 10x efficiency

Resources & Further Learning

Microsoft Word

- support.microsoft.com/word
- Microsoft Learn — Word Training
- YouTube: Microsoft 365 Official Channel
- LinkedIn Learning: Advanced Word

Microsoft Excel

- support.microsoft.com/excel
- ExcelJet.net — Formula Reference
- Chandoo.org — Excel Charting Guides
- Microsoft Learn — Excel Fundamentals

AI & Copilot

- microsoft.com/copilot
- learn.microsoft.com/copilot
- Microsoft Copilot for Education (M365)
- Bing Chat for document assistance

Practice Datasets

- Kaggle.com — Free CSV datasets
- data.gov.in — Indian Government data
- Google Dataset Search
- NIELIT GitHub resources

Thank you!
Questions Welcome

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